

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First / Second Semester 2024-2025

Comprehensive Examination
(EC-3 Regular)

Course No. : AIMLCZG565/DSECLZG565
Course Title : Machine Learning
Nature of Exam : Open Book
Weightage : 40%
Duration : 2 Hours
Date of Exam : **** (FN/AN)

No. of Pages	= s
No. of Questions	=

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Please Note:

A **maximum of 12 main questions** with Q.Nos. 1 to N.

Q.1. [3 Marks]

You are the **Lead ML Engineer** for a global bank building a regression model to predict **Annual Default Risk** (a continuous value between **0 and 1**) for corporate clients.

Key details of the problem:

- Dataset size: **60,000 samples**
- Number of features: **500**
- Domain experts estimate that **only 10–15 features are truly influential**
- Many features are **highly correlated**
- The **Regulatory Compliance Office** mandates that:
 - The model must be **auditable**
 - The contribution of each feature to a prediction must be **explainable**

Your team evaluates multiple models and observes:

- A **Lasso-based linear model** achieves an RMSE of **0.081**
- A **complex Ensemble Regressor** (Random Forest / Gradient Boosting) achieves an RMSE of **0.062**

Answer the following:

- (a) Which regression model is most appropriate for this task, given the data size and domain constraints? Justify your choice.
- (b) Despite the lower RMSE, explain **two specific disadvantages** of deploying the Ensemble Regressor in this regulatory setting.

Solution:

(a) Recommended model (*1 mark*)

Award full marks if **any two** of the following are mentioned:

- **Lasso Regression (L1-regularized linear regression)**
- Encourages **sparse solutions**
- Automatically sets **irrelevant coefficients to zero**
- Suitable when **few features are expected to be important**
- Linear structure supports **auditability**

(b) Disadvantages of the Ensemble model (2 marks – 1 mark each)

Award **1 mark per valid disadvantage**:

- **Low interpretability**
 - No clear, single equation explaining predictions
 - Hard to trace feature-level contribution
- **Poor auditability**
 - Difficult to justify decisions to regulators
 - Complex internal logic (many trees / interactions)
- **Regulatory risk**
 - Hard to verify compliance with fairness and decision rules
- **Limited explainability for end users**
 - Feature importance is approximate, not causal

Q.2.

[6 Marks]

A fintech company is developing a **risk profiling system** to classify customers into **Low Risk (R1)** or **High Risk (R2)** categories using a **distance-weighted KNN classifier**. Each customer record contains a mix of numerical, categorical, and ordered attributes. The model assigns higher importance to customers that are more similar to the query customer, using a distance-based kernel.

The historical dataset and a new query customer **Q** are shown below.

Customer	Age	Monthly Spend (₹)	Employment	City Tier	Risk
A0	35	40,000	Salaried	Tier-3	R2
A1	45	70,000	Salaried	Tier-2	R2
A2	30	38,000	Self-Employed	Tier-3	R1
A3	50	90,000	Salaried	Tier-1	R2
A4	28	42,000	Self-Employed	Tier-2	R1
Q (Query)	32	41,000	Self-Employed	Tier-3	?

The classifier uses a **distance-weighted KNN** approach with kernel

$$K(d)=1/ d^2$$

and considers **all training instances**. All attributes are treated as equally important.

Answer the following:

(a) Identify an appropriate distance measure for this dataset and compute the **distance-weighted class score** for **R1** and **R2**, then determine the predicted risk class for **Q**. [3 Marks]

(b) Even though **R2 has more training samples than R1**, the distance-weighted KNN predicts **R1**. Give **one precise reason**, based on the computed distances, for why this happens. [1 Mark]

(c) Recompute the distances and the distance-weighted class scores assuming **City Tier** is treated as a nominal attribute instead of an ordinal attribute. Based on this recomputation, **does the final predicted class for Q change or remain the same?** Compute the difference in the prediction score for the assigned risk class. [2 Marks]

Solution:

Part (a): Distance Computation and Classification [3 Marks]

Step 1: Identify distance measure - Mixed numeric, nominal, and ordinal attributes → Gower distance (✓ 0.5 mark)

Step 2: Normalize numeric attributes (Age, Monthly Spend) - Min-max normalization applied correctly (✓ 0.5 mark)

Step 3: Attribute-wise distances

- Numeric → absolute normalized difference
- Employment → 0 (match), 1 (mismatch)
- City Tier → ordinal distance using rank normalization (✓ 0.5 mark)

Step 4: Gower distance computation

$$d = (d_age + d_spend + d_employment + d_city) / 4 \text{ (✓ 0.5 mark)}$$

Step 5: Distance-weighted voting

- Weight = $1/d^2$ (0.5 marks)

Customer	Risk	d_age	d_spend	d_emp	d_city	Gower d	Weight (1/d ²)
A2	R1	0.0909	0.0577	0	0.0	0.03715	724.56
A4	R1	0.1818	0.0192	0	0.5	0.17526	32.56
A0	R2	0.1364	0.0192	1	0.0	0.28890	11.98
A1	R2	0.5909	0.5577	1	0.5	0.66215	2.28
A3	R2	0.8182	0.9423	1	1.0	0.94012	1.13

Step 5: Class-wise weight sums

- **R1 score** = $w(A2) + w(A4) = 724.56 + 32.56 = 757.12$
R2 score = $w(A0) + w(A1) + w(A3) = 11.98 + 2.28 + 1.13 = 15.39$

Prediction (weighted KNN): R1 (since $757.12 > 15.39$)

- class weights and Final prediction: R1 (✓ 0.5 marks)

Part (b): Why R1 wins despite fewer samples [1 Marks]

- Distance-weighted KNN emphasizes similarity, not count (✓ 0.5 mark)
 - Closest R1 neighbor has very small distance → very large weight (✓ 0.5 mark)
- Or any other valid point

Part (c): Re-computation with City Tier as Nominal [2 Marks]

City Tier treated as nominal: 0 if same, 1 if different

Recompute distances (0.5 Marks)

Customer	Risk	Old d_city	New d_city	New Gower d	New weight
A2	R1	0.0	0.0	0.03715	724.56
A4	R1	0.5	1.0	0.30026	11.09
A0	R2	0.0	0.0	0.28890	11.98
A1	R2	0.5	1.0	0.78715	1.61
A3	R2	1.0	1.0	0.94012	1.13

- Distances to Tier-2 customers increase

Recompute weights (0.5 marks)

Final Outcome:

- Prediction remains R1 (0.5 marks)
- Margin reduces $757.12 - 735.65 = 21.47$ (0.5 marks)

Q.3.

[8 Marks]

An AdaBoost classifier is being trained on 4 samples. At iteration 3, the sample weights are:

Sample	1	2	3	4
Weight (w_i)	0.1	0.3	0.15	0.45
True Label (y_i)	+1	+1	-1	-1

A weak learner $h_3(x)$ is trained and makes the following predictions:

Sample	1	2	3	4
Prediction $h_3(x_i)$	+1	-1	-1	-1

- Identify which samples are misclassified by h_3 . **[0.5 mark]**
- Calculate the weighted error rate $\epsilon_3 = \sum w_i$ for all misclassified samples. **[1 marks]**
- Calculate the learner weight $\alpha_3 = \frac{1}{2} \ln((1 - \epsilon_3) / \epsilon_3)$. **[2 marks]**
- Which sample has the highest weight? What does this tell us about AdaBoost's learning strategy? **[2.5 marks]**
- State one advantage and one disadvantage of AdaBoost compared to a single decision tree. **[2 mark]**

Solution:

(a) Misclassified Samples [0.5 mark]

Compare each prediction with true label:

Sample 1: True = +1, Predicted = +1 → ✓ Correct

Sample 2: True = +1, Predicted = -1 → ✗ **Misclassified**

Sample 3: True = -1, Predicted = -1 → ✓ Correct

Sample 4: True = -1, Predicted = -1 → ✓ Correct

Answer: Only Sample 2 is misclassified

(b) Weighted Error Rate [1 marks]

Formula: ϵ = sum of weights of misclassified samples

Step 1: Identify misclassified samples and their weights

Misclassified: Sample 2 only

Weight of Sample 2: $w_2 = 0.3$

Step 2: Sum the weights

$$\epsilon_3 = w_2 = 0.3$$

$$\epsilon_3 = 0.3$$

(c) Learner Weight Calculation [2 marks]

Formula: $\alpha = \frac{1}{2} \times \ln((1 - \epsilon) / \epsilon)$

Step 1: Calculate $(1 - \epsilon_3)$

$$1 - \epsilon_3 = 1 - 0.3 = 0.7$$

Step 2: Calculate the ratio $(1 - \epsilon_3) / \epsilon_3$

$$(1 - \epsilon_3) / \epsilon_3 = 0.7 / 0.3 = 2.333$$

Step 3: Take natural logarithm

$$\ln(2.333) \approx 0.847$$

Step 4: Multiply by $\frac{1}{2}$

$$\alpha_3 = \frac{1}{2} \times 0.847 = 0.424$$

$$\alpha_3 \approx 0.42$$

(d) High Weight Explanation [2.5 marks]

Sample 4 has the highest weight because it was likely **misclassified by previous weak learners** (iterations 1 and 2). AdaBoost increases weights on misclassified samples after each iteration. This is AdaBoost's key strategy: it forces subsequent learners to focus on the "hard" examples that earlier learners got wrong, ensuring the ensemble progressively improves on difficult cases.

(e) AdaBoost vs Single Tree [2 mark]

Advantage: Higher accuracy — combining multiple weak learners typically outperforms a single tree by reducing bias.

Disadvantage: Sensitive to noisy data and outliers — AdaBoost keeps increasing weights on hard examples, which may be noise, leading to overfitting.

Q.4.

[6 Marks]

A customer-support analytics team is building an automated system to classify customer emails into two categories: Complaint (C) and Inquiry (I).

The system represents each email using a bag-of-words model, where only the frequency of specific keywords is considered. Based on historical data, the following vocabulary is used:

$$V = \{\text{delay, refund, order, price}\}$$

Training statistics are summarized below:

Word counts in Complaint (C): delay=8, refund=6, order=2, price=1

Word counts in Inquiry (I): delay=1, refund=1, order=6, price=7

Additional information:

Number of training emails in class C = 5

Number of training emails in class I = 5

Total word count in class C = 17

Total word count in class I = 15

A new incoming email contains the words:

"delay refund refund"

The classifier uses Multinomial Naive Bayes with Laplace smoothing ($\alpha = 1$).

Answer the following:

- (a) Compute the prior probabilities for classes C and I. [1 mark]
- (b) Compute the Laplace-smoothed conditional probabilities for words delay and refund under both classes. [3 marks]
- (c) Compute the posterior probabilities (up to proportionality) and determine the predicted class. [2 marks]

Solution:

(a) Prior Probabilities [1 mark]

$$P(C) = 5 / 10 = 0.5 \quad (\checkmark 0.5 \text{ mark})$$

$$P(I) = 5 / 10 = 0.5 \quad (\checkmark 0.5 \text{ mark})$$

(b) Laplace-Smoothed Conditional Probabilities [3 marks]

Vocabulary size $|V| = 4$

Laplace smoothing formula:

$$P(w|y) = (\text{count}(w,y) + 1) / (\text{total words in } y + |V|)$$

For class C (total words = 17):

$$P(\text{delay}|C) = (8+1)/(17+4) = 9/21 = 0.429 \quad (\checkmark 0.75)$$

$$P(\text{refund}|C) = (6+1)/(17+4) = 7/21 = 0.333 \quad (\checkmark 0.75)$$

For class I (total words = 15):

$$P(\text{delay}|I) = (1+1)/(15+4) = 2/19 = 0.105 \quad (\checkmark 0.75)$$

$$P(\text{refund}|I) = (1+1)/(15+4) = 2/19 = 0.105 \quad (\checkmark 0.75)$$

(c) Posterior Computation and Classification [2 marks]

Email: "delay refund refund"

Using Multinomial NB:

$$P(y|x) \propto P(y) \times \prod P(w|y)^{\text{count}(w)}$$

For class C:

$$P(C|x) \propto 0.5 \times (0.429)^1 \times (0.333)^2 \approx 0.0238 \quad (\checkmark 1 \text{ mark})$$

For class I:

$$P(I|x) \propto 0.5 \times (0.105)^1 \times (0.105)^2 \approx 0.00058 \quad (\checkmark 1 \text{ mark})$$

Since $P(C|x) \gg P(I|x)$, the predicted class is Complaint (C).

Q.5.

[8 Marks]

A bank is building an SVM classifier to predict loan defaults. They have trained two models:

- Model A: Linear SVM with $C = 0.1$, achieved 85% training accuracy, 84% test accuracy
- Model B: Polynomial kernel SVM with $C = 100$, degree=6, achieved 98% training accuracy, 72% test accuracy

(a) Which model is overfitting? Explain what the parameters C and degree do, and why the chosen values in Model B lead to this problem. Also comment on what this suggests about the **linear vs non-linear separability** of the data. [3 marks]

(b) The bank uses Model A with weight vector $w = [0.5, -0.3]^T$ and bias $b = 0.2$.

A new applicant has features $x = [2, 1]^T$ (representing income and debt ratio).

Calculate $f(x) = w^T x + b$.

Determine whether this applicant will default (positive class) or not default (negative class). **[2 marks]**

(c) Calculate the geometric margin of the classifier. What does a larger margin indicate about the classifier's confidence? **[2 marks]**

(d) Why might the bank prefer soft-margin SVM over hard-margin SVM for this problem? **[1 mark]**

SOLUTION

(a) Overfitting Diagnosis [3 marks]

Model B is overfitting — high training accuracy (98%) but much lower test accuracy (72%) indicates the model memorized training data but doesn't generalize. [1 mark]

Parameter C: Controls the penalty for misclassification. High C (=100) forces the model to classify almost all training points correctly, even if they are noise or outliers. This creates a complex decision boundary. [0.5 mark]

Degree: Controls the degree of the polynomial kernel function. [0.5 mark]

Separability insight: [1 mark]

- Model A's similar train/test accuracy suggests the data is **approximately linearly separable (with some overlap)**
- Model B's behavior indicates **non-linear patterns exist**

(b) Classification Calculation [2 marks]

Given:

$w = [0.5, -0.3]^T$ (weight vector)

$b = 0.2$ (bias)

$x = [2, 1]^T$ (new applicant's features)

Step 1: Calculate $w^T x$ (dot product)

$$w^T x = (w_1 \times x_1) + (w_2 \times x_2)$$

$$w^T x = (0.5 \times 2) + (-0.3 \times 1)$$

$$w^T x = 1.0 + (-0.3)$$

$$w^T x = 0.7$$

Step 2: Calculate $f(x) = w^T x + b$

$$f(x) = 0.7 + 0.2$$

$$f(x) = 0.9$$

Step 3: Determine classification

Since $f(x) = 0.9 > 0$, the prediction is **positive class (DEFAULT)**

(c) Geometric Margin [2 marks]

Step 1: Calculate $\|w\|$ (magnitude of weight vector)

$$\|w\| = \sqrt{(w_1^2 + w_2^2)}$$

$$\|w\| = \sqrt{(0.5^2 + (-0.3)^2)}$$

$$\|w\| = \sqrt{(0.25 + 0.09)}$$

$$\|w\| = \sqrt{0.34}$$

$$\|w\| \approx 0.583$$

Step 2: Calculate geometric margin

$$\text{Geometric margin} = 1 / \|w\|$$

$$\text{Geometric margin} = 1 / 0.583$$

Geometric margin ≈ 1.715

Interpretation: A larger margin means the decision boundary is farther from the nearest training points, indicating higher confidence and better expected generalization to new data.

(d) Soft vs Hard Margin [1 mark]

Loan data likely has overlapping classes (some defaulters look similar to non-defaulters). Hard-margin SVM requires perfect separation which may be impossible. Soft-margin allows some misclassifications using slack variables, finding a practical boundary that balances margin size with training errors.

Q.6.

[6 Marks]

A retail analytics team is clustering customers based on two features:

x1: Average monthly spend

x2: Visit frequency

The following 2-D dataset is obtained:

Point	x ₁	x ₂
A	1	2
B	2	1
C	2	3
D	6	5
E	7	6
F	8	5

a) The analytics team decides to partition the given 2-D customer data into **two groups** using a **centroid-based clustering approach**. At the start of the procedure, two representative points in the feature space are selected as the initial cluster centers, located at (1,2) and (8,5).

Using these initial centers as references, determine how the data points would be grouped after the **first complete update cycle** of the K-Means algorithm. Provide a step-by-step explanation of:

- how proximity to the current centers influences membership
- how the representative locations of the two groups change once the memberships are determined

Clearly state the resulting groups and the updated representative points obtained at the end of this cycle. **[3 marks]**

b) Now assume the same data is modeled using a **Gaussian Mixture Model (GMM)** with two components having:

- Equal mixing weights
- Spherical covariance matrices
- Means initialized to the centroids obtained in part (a)

Explain **one key difference** between cluster assignments in K-Means and responsibilities in GMM, using **one specific point from the dataset** to justify your answer. **[2 marks]**

c) State **one data characteristic** under which GMM would produce **better clusters than K-Means** for this dataset, and explain why. **[1 mark]**

Solution:

(a) K-Means – One Iteration [3 marks]

Distance-based assignment

- Points A, B, C are closer to C1
- Points D, E, F are closer to C2

Cluster 1: A(1,2), B(2,1), C(2,3)

Cluster 2: D(6,5), E(7,6), F(8,5)

Updated centroids

Cluster 1:

$$\mu_1 = (1+2+2, 2+1+3) = (1.67, 2) \quad \mu_1 = \left(\frac{1+2+2}{3}, \frac{2+1+3}{3} \right) = (1.67, 2)$$

Cluster 2:

$$\mu_2 = (6+7+8, 5+6+5) = (7, 5.33) \quad \mu_2 = \left(\frac{6+7+8}{3}, \frac{5+6+5}{3} \right) = (7, 5.33)$$

(b) K-Means vs GMM Assignment [2 marks]

Key difference:

- **K-Means** assigns each point **hardly** to exactly one cluster
- **GMM** assigns **soft responsibilities** (probabilities) to each cluster

Example:

Point C (2,3) is clearly in Cluster 1 under K-Means.

In GMM, it may still have a **non-zero probability** of belonging to Cluster 2 depending on covariance and distance.

(c) When GMM is Better [1 mark]

If clusters are **elliptical or overlapping**, GMM performs better because it:

- Models **cluster shape via covariance**
- Accounts for **uncertainty in assignment**

Q.7. [3 Marks]

A fintech company is building machine learning models to predict loan default using the same two features:

x1: Normalized income

x2: Normalized credit utilization

Three different models are trained on the same dataset:

- Model A: Linear Regression (used with a threshold to classify)
- Model B: Logistic Regression
- Model C: Decision Tree (maximum depth = 3)

During evaluation, the following observations are made:

- Model A sometimes outputs values outside the range [0, 1]
- Model B outputs values close to 0 or 1 for extreme values of x_1
- Model C shows sudden changes in predicted class for small changes in x_2

Answer the following:

- (a) Explain why Model A can produce predictions outside the range [0,1] while Model B cannot, even though both models use linear combinations of the same input features. [1 mark]
- (b) As the size of the training dataset decreases, which of the three models is most likely to overfit, and why? Your answer must explicitly refer to how each model represents its decision boundary. [1 mark]
- (c) A regulatory auditor asks which model's prediction behavior will change most abruptly for a very small change in input features, and why this is undesirable in high-stakes decision systems. Identify the model and justify your answer. [1 mark]

Solution:

- (a) Mathematical interpretation (1 mark):

Linear Regression computes predictions as $\hat{y} = w^T x + b$, which is an unbounded linear function. As a result, outputs can be less than 0 or greater than 1.

Logistic Regression applies a sigmoid function to the linear score, mapping all real-valued inputs to the interval (0,1), making probabilistic interpretation possible.

Full mark for **clearly stating bounded vs unbounded behavior**.

- (b) Overfitting and decision boundaries (1 mark):

The Decision Tree is most likely to overfit when training data is limited because it learns flexible, piecewise-constant decision boundaries using axis-aligned splits.

Linear and Logistic Regression impose global linear decision boundaries, providing stronger inductive bias and better generalization in low-data settings.

Full mark for **linking overfitting to model flexibility and boundary shape**.

- (c) Robustness and abrupt prediction changes (1 mark):

The Decision Tree shows the most abrupt changes in prediction. Small variations in input values can move a data point across a split threshold, causing a sudden change in predicted class.

Such discontinuous behavior is undesirable in high-stakes systems because it reduces stability, fairness, and user trust.

Full mark for **identifying discontinuity and its implications**.
